A Multilayer Resilience Assessment of Power Distribution Systems with Reliability Models, Service Restoration, and Dynamic Bayesian Networks

Michel Bessani, Henrique O. Caetano, Luiz Desuó N., Matheus S. S. Fogliatto and Carlos D. Maciel

Abstract Electrical energy is fundamental for contemporary society since failures directly impact other critical infrastructures such as water and gas distribution, hospitals, or banking services. Consequently, resilience, which is the capability of a system to handle high-impact low probability events, is a crucial aspect of such systems. Efficient resilience assessment methods are essential to achieving high-performance, resilient energy systems. This chapter introduces a multilayer method to address several factors of power distribution systems' resilience. Reliability regressions model the failures' instant and duration given a weather scenario, a dynamic Bayesian network (DBN) models how probabilities of failure propagate on the system's physical connections, and a service restoration through switching operations, and field crew routing is obtained through an optimization algorithm for a given set of failures. Information related to these factors has the potential to be structured in a layered manner for a better understanding of the dynamic interaction among different information like weather, routes, power grid, and historical events logs. The ability to model these relationships enables the inference of the system resilience

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for different inputs during analysis. Resilience can also be inferred by considering the uncertainties associated with these layers due to DBN's nature. A case study is presented to show the efficacy of this procedure. The findings showed its ability to evaluate the resilience of power distribution systems in the face of uncertainty and the considered aspects for different weather scenarios.

Keywords: Resilience Assessment, Dynamic Bayesian Networks, Reliability, Service Restoration, Power Distribution Systems

1 Introduction

The reliance on electricity in society is growing at an escalating rate [1]. This dependency is further intensified by the digitalization of existing services and processes and the progress of cyber-physical systems, which integrate cyber components to enhance or provide new functions to physical systems [2]. This interdependence is expected to intensify further in the near future with the advent of Industry 5.0 and Society 5.0 [3]. Furthermore, as stated in the 2023 report of the International Panel on Climate Change (IPCC) [4], there is an anticipated rise in both the frequency and intensity of extreme climate events. Extreme events typically burden Critical Infrastructure (CIs), including water and gas distribution, hospitals, communication, and power systems. This leads to prolonged impairment of these systems, resulting in service interruptions, economic losses, and negative impacts on societal well-being.

Resilience becomes the focal point when examining how CIs react to significant failures on a broad scale and in the face of various threats. It involves the ability of a CI to prepare for, withstand, and recover from significant disruptions. Resilience is linked to the ability to prepare for and adapt to changing conditions and inherently involves a temporal element. This temporal dimension is essential for understanding how CIs can withstand and recover from disruptions, whether deliberate attacks, accidents, or natural disasters [5]. Therefore, CIs' resilience can be viewed as a dynamic and ongoing characteristic that unfolds over time, encompassing preparedness, response, and recovery efforts to ensure the continuity of essential functions.

The definition of resilience that we adopt is derived from The National Academy of Sciences, Engineering, and Medicine of the United States [6]. According to this definition, resilience is the capacity to effectively prepare for, absorb, recover from, or adapt to current and potential adverse events. Within this framework, it is crucial to emphasize three fundamental components of resilience:

1. Temporal aspect is key for resilience assessment, especially the periods before, during, and after a disruptive event [7];
2. Resilience is scenario dependent, meaning that internal (operational characteristics) and external (attacks or weather variables) factors affect the failure occurrence and response of the system during a disruptive event, and consequently, its resilience [8];

3. Uncertainties will always be presented when assessing the resilience of complex systems. There must be a proper manner to evaluate the performance of a typical power system in the face of different scenarios [9].

Research has mostly focused on the resilience of several critical infrastructures, including water transportation [10], railway networks [11], and others [12]. In the realm of power systems, the ability to operate resiliently is of utmost importance. This is because other important systems and services rely on electricity, making it imperative for power distribution systems (DS) to prioritize resilience. Notably, the bulk of failures occur in power distribution systems [13]. The research on the resilience of Distribution Systems is interdisciplinary and primarily concentrates on evaluating or improving resilience. DSs resilience assessment can be conducted through various strategies, including historical data analysis, analytical methods, and probabilistic approaches such as stochastic simulation and inference methods.

In addition to considering the advantages and disadvantages of these methodologies, it is important to acknowledge the inherent uncertainty in the resilience analysis of complex systems. These systems, such as power systems and other elaborate engineered systems, often display intricate interdependencies. These complex systems function on multiple levels of knowledge and factors, as will be explained in Section 3. As a result, recent research efforts have shifted towards a probabilistic approach, highlighting the importance of systematically considering and evaluating uncertainties in resilience assessment. It is crucial to adopt probabilistic frameworks because they give decision-making power to CI operators and allow for the use of advanced probabilistic methodologies to improve our understanding of the resilience of complex systems. Consequently, this facilitates the implementation of more efficient measures to reduce the impact of risks [9], thereby promoting enhanced ability to recover from adverse events in different essential areas.

The uncertainties related to the resilience of DS can arise from various causes. High-impact, low-probability (HILP) events, such as storms, typhoons, earthquakes, floods, fires, droughts, and heatwaves, are external influences on the system. The uncertainty lies in their incidence, length, and intensities. These incidents have an impact on the DSs as well as other urban infrastructures. Furthermore, distinct operational uncertainties in the DSs are associated with the intermittent nature of demand and generation, the occurrence of failures, the ability to restore service, the availability of field crews, and the duration of repairs. These uncertainties are internal factors.

Numerous studies in the literature have addressed the issue of evaluating resilience using a probabilistic approach. The study conducted in [14] examines the assessment of uncertain factors using the Graphical Evaluation and Review Technique. Meanwhile, the authors in [15] utilize a Markov Process to evaluate the failure and repair process of CIs. The study conducted by [16] applies random multi-hazards and considers a stochastic recovery process to evaluate temporal resilience. The study conducted by [17] evaluated the resilience of DSs using a Monte Carlo simulation. This simulation incorporated reliability models that considered both failure and repair processes, which were, in turn, affected by weather conditions. In the

article [18], a novel approach based on dynamic Bayesian Networks was introduced to model and evaluate the resilience of DSs.

The studies exploiting power system resilience as a multi-layered model have been developed in the context of analyzing the resilience of modern power systems. The resilience of power systems is a complex and multifaceted concept involving various interconnected factors. For instance, a study on the resilience of modern power systems from a complex network perspective investigates the modeling, evaluation, and enhancement of structural resilience in power systems based on a complex network framework [19]. Additionally, a multi-layered Directed Acyclic Graph (DAG) model has been used for resilience formulation in the context of bulk power systems, emphasizing the interconnected layers of the system [20]. Furthermore, quantifying resilience as a key performance indicator is important for power systems, and it involves analyzing existing power systems and identifying resilience improvements in future power systems [21]. These examples demonstrate the application of a multi-layered approach in understanding and enhancing the resilience of power systems.

This chapter introduces a multilayer resilience assessment approach that utilizes a dynamic Bayesian Network (DBN) to model the components' failures and repair events. Evidence propagation is employed to facilitate the resilience assessment process. In addition, the resilience evaluation takes into account the following external and internal factors: *i)* Weather conditions that impact the frequency of breakdowns and the time taken for repairs; *ii*) The frequency of failures, which is influenced by the likelihood of component failure and is directly influenced by weather conditions; *iii*) The process of restoring service, which aims to minimize the impact of failures on the system's performance through switching operations; and *iv*) The actions taken by field crews to mitigate the effects of failures, such as using manual switches to improve service restoration and repairing the failed components.

The relation between these factors and their impact on the DS components is depicted in a layered framework. This schema enables the inference of the system's resilience from various scenario inputs by modeling the interaction between these elements and the system's components during the resilience assessment. Furthermore, the DBN inherently represents the ambiguity that arises during evaluating resilience. The subsequent portion of this chapter is structured in the following manner: Section 2 explains the basic principles of DBN. Section 3 outlines the multilayer resilience assessment approach and how it arranges the models and algorithms for the various factors, then presents the reliability models for failure occurrence instant and duration, the service restoration algorithm, and the resilience assessment through DBN. Section 4 demonstrates the application of the multilayer approach in a case study. Finally, Section 5 provides the concluding remarks for this chapter.

2 Dinamic Bayesian Networks

Bayesian Networks (BNs) are a valuable tool for knowledge representation, particularly when dealing with complex systems [22]. A BN is essentially built upon a Directed Acyclic Graph (DAG), wherein each node in the graph represents a random variable. The edges, depicted as arrows connecting these nodes, signify the statistical dependencies between these variables. These arrows point from parent nodes to child nodes, establishing the flow of influence within the network. Nodes that lack any incoming arrows are referred to as root nodes and are assigned marginal probabilities. For the remaining nodes, conditional probabilities come into play. These probabilities capture the likelihood of each child node's value, considering the values of all its parent nodes, and are stored in a Conditional Probability Table (CPT). In essence, a Bayesian Network acts as a structured framework to map out relationships, dependencies, and probabilities, facilitating a deeper understanding of complex systems [22].

In a BN, nodes correspond to random variables, each with assigned probabilities representing possible states conditioned on their parent nodes. Evidence propagation is an important technique for exploring BNs' inference capabilities for assigning probabilities to nodes under specific conditions. In the context of BNs, assigning a specific value to a node as evidence (denoted as "e") differs from incorporating a new set of observations (denoted as "E") in terms of their scope and impact on the network. When referring to a new set of observations (E), it implies a collection of observed values, i.e., a random sample of the random variables modeled by the BN. These observations are then used to update the probabilities of other nodes through Bayesian inference, influencing the entire network.

Conversely, when we assign a specific value as evidence (e) to a particular node, we take a more focused approach. This involves fixing the value of a specific node in the network and asserting the truth or occurrence of the observed evidence. This localized evidence assignment indirectly affects the probabilities of other nodes in the network through their dependencies on the evidence node.

To further illustrate the presented concepts, Figure 11 (a) depicts a typical BN, adapted from an example in [23]. This BN consists of four nodes: Cloudy, Sprinkler, Rain, and Wet Grass. Essentially, the weather condition (whether it's cloudy or not) influences not only the presence of Rain but also whether the Sprinkler is on or off. Moreover, both the Sprinkler being on and the presence of rain impact whether the grass is wet. The process of defining the BN's structure, including its nodes and edges, encodes the presence or absence of dependencies between the variables (nodes), while the CPTs precisely define the strength of these dependencies.

On the other hand, Figure 11 demonstrates how the process of evidence propagation affects the BN. Initially, a subset of nodes is selected to apply the evidence (in this case, only the node Wet Grass). The process involves setting each node with evidence to a specific state. In our example, the Wet Grass node is set to "True" and we aim to understand how this evidence will impact the probabilities of other nodes. While the BN's structure remains unchanged, the CPTs must be recalculated

(represented by question marks in the figure) for all nodes except the one where the evidence is being evaluated.

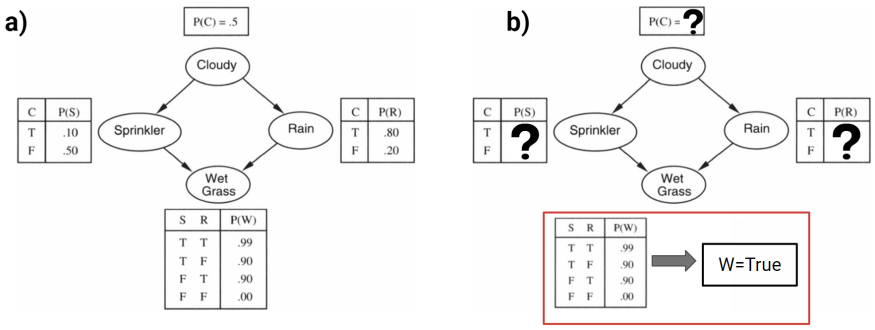


Fig. 1: This example demonstrates a typical Bayesian Network, as shown in a), with its structure (nodes and edges) and CPTs. In Figure b), node "Wet Grass" is fixed in a given state, and all other probabilities must be updated accordingly.

DBNs are an extension of BNs, where the temporal aspect is added. The DBN is considered over a given time window consisting of T time instants, where for each time slice t ($t = 0, ..., T$), there exists a DAG $\mathbb{G}_t = (V_t, E_t)$, where V_t is the set of nodes and E_t is the set of edges. Finally, for $t = 1, ..., T$, there is a set $E_{t-1,t}$ of transition edges, which make the connection between V_{t-1} and V_t. Let $\mathbb{G} = \{\mathbb{G}_0, ..., \mathbb{G}_t\}$, and P be the set of CPTs defined for each node, then the set $(\mathbb{G}, P)$ forms a Dynamic Bayesian Network [24]. A DBN allows keeping the mentioned advantages of BNs, while also adding the temporal aspect by considering the random variables over a set of discrete time slices, with a static BN for each one [22]. In this manner, the edges connecting elements in a DBN can be divided into two types:

1. Intratemporal edges: edges that connect nodes in the same time slice;
2. Intertemporal edges: edges that connect the same or distinct nodes in different time slices.

To delve deeper into the distinctive features of DBNs compared to static BNs, Figure 2 illustrates a prototypical DBN comprising three nodes (A, B, and C) across three temporal slices ($t \in \{0, 1, 2\}$). In this depiction, intertemporal edges are highlighted in green, signifying dependencies across different time points, while intratemporal edges are depicted in yellow. It is noteworthy that intratemporal edges may connect the same node across different time slices, as exemplified by the connection between nodes A_0 and A_1, or between distinct nodes, as seen between nodes C_0 and B_1. Notably, while the set of intratemporal edges typically remains constant across time slices [22], CPTs may vary for the same node across different time points. For instance, the CPT for nodes A_0 and A_1 could differ, reflecting temporal changes in node probabilities. This dual mechanism enables DBNs to capture temporal dynamics: firstly, by modeling the interactions between nodes across time slices

via intratemporal edges, and secondly, by adjusting CPTs over time to encapsulate temporal evolution. Additionally, it is crucial to emphasize that the process of evidence propagation seamlessly applies to DBNs without necessitating any additional modifications [22].

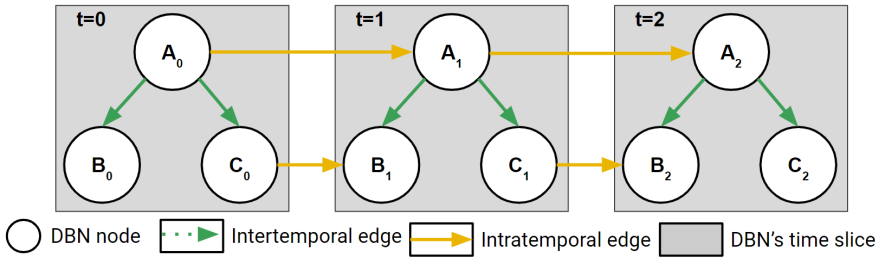


Fig. 2: This example demonstrates a typical Dynamic Bayesian Network, with its structure (nodes and edges) and CPTs. Each time slice represents static nodes for a particular time, and edges between time slices represent the temporal dependence between variables

3 Multilayer Resilience Assessment

This Section discusses the proposed multilayer resilience assessment approach. Firstly, the multilayer structure that combines the considered knowledge and factors, the DBN and Failure duration and occurrence models, and the algorithms to perform evidence propagation, service restoration, and resilience estimation is described in Subsection 3.1. Then, the information, models, and algorithms for the considered factors are presented. Respectively, failures occurrence and duration reliability models that reflect the effect of weather scenarios on failures and repair probabilities in Subsection 3.2, service restoration through manual switches operation and field teams routing in Subsection 3.3, and the resilience assessment using DBN in Section 3.4.

3.1 Multilayer Arrangement

The multilayer arrangement of the factors, knowledge, models, and algorithms is depicted in Figure 3. The black boxes describe the knowledge necessary to execute our approach, which is the data of the DS used to define the DBN structure and to execute service restoration, and the historical data of failures for such DS used to learn the failure occurrence and duration models. The weather scenario gray box is

the input used to probabilistically assess the resilience. A given weather scenario results in a failure scenario where service restoration and evidence propagation are executed to allow resilience estimation through MAP performance inference. The white boxes are the models learned from data: the DBN structure learned from the DS data, and the DBN CPTs based on the failure occurrence and duration models which are also influenced by the given weather scenario. The ellipses represent the algorithms used to process the failure scenario sampled from the DBN CPT, and the circle is the resilience estimation based on the MAP performance inference.

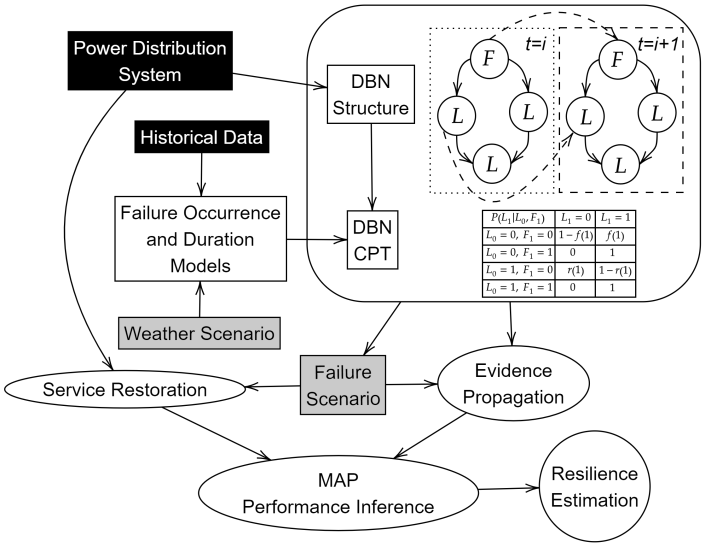


Fig. 3: Multilayer structure that combines knowledge and factors on learned models and runs algorithms to process a given weather scenario and perform resilience estimation.

3.2 Failures Occurrence and Duration Models

The power system is divided into three subsystems: Generation, Transmission, and Distribution. Generation is the part of the system responsible for generating electricity. Several generation methods exist, such as thermal power plants, hydroelectric plants, wind and solar panel farms, and nuclear plants. Generation failures generally occur due to equipment failures or operational errors. However, especially in renewable sources, it is not uncommon for operations to be suspended due to a lack of the base resource, with an empty water reservoir in hydroelectric plants.

Transmission is the part that connects generation to distribution systems. If, on the one hand, generating sources such as thermoelectric plants are generally located close to large urban zones, renewable sources are generally further away. Considering this, the structure of the transmission system varies greatly. The function of transmission systems is to transport electricity under conditions that avoid losses as much as possible. To transport electricity in these conditions, the structure and equipment of the transmission networks are robust. Another point is that as they are not located within large urban zones, their contact with human elements and nature is much less, reducing the risk of accidents that could damage part of the system and generate failures. However, when these disruptive events occur, the impact is significantly large, with large areas becoming de-energized.

Distribution is the system's last part and is responsible for transporting electricity and lowering the electrical quantities to ideal values for use by end consumers. It is the part that has the most number of equipment, and its structure is located within urban areas. Therefore, it is the power system part most subject to failures, which can be operational, such as equipment defects, and related to external factors, such as accidents with vehicles, animals, vegetation, or events related to atmospheric events, such as wind, lightning, and rain.

Failures are an inevitable problem in Power Systems. Any interruption in the electricity supply is considered a failure. A large part of their structure is located in open places and exposed to the environment, which ends up facilitating the occurrence of accidents. Accidents involving vehicles crashing into parts of the system, vegetation touching or falling on equipment, and atmospheric events that cause damage to various parts of the network are examples [25, 26, 27, 28] of uncertainty factors that generate failures in the electrical system.

Due to the direct impact on end consumers, such as commerce, industry, and the general population, distribution networks are currently the area with the most significant emphasis in academic studies. Investigating improvements in the areas of system operation and planning, such as finding the best system configuration and the best positioning for protective equipment, as well as understanding the problems faced by calculating reliability indices, and identifying the most serious threats, are some of the examples of study areas in distribution systems [29, 30, 31].

The most common way to evaluate a Power System's behavior during failures is through reliability indices [2, 32, 33, 34], calculated with the quantity or duration of failures experienced by a distribution system and the number of customers served by this system. The System Average Interruption Frequency Index (SAIFI) and System Average Interruption Duration Index (SAIDI) were the usual metrics to measure the failure impacts on power systems. According to local legislation in each region, system administrators are subject to fines depending on the values of these metrics. Eq. 1 and 2 show how SAIFI and SAIDI are calculated.

$$SAIFI = \frac{Total\ Number\ of\ Interruptions}{Number\ of\ Customers} \tag{1}$$

$$SAIDI = \frac{Total\ Duration\ of\ Interruptions}{Number\ of\ Customers} \tag{2}$$

Although weather events are not the most common cause of failures, their impact is considerable on the duration of these disruptive events and the number of customers affected. Extreme atmospheric events, such as storms, can result in regions without electricity for days or weeks. Furthermore, they influence other types of problems, such as increasing the rate of accidents involving network components. Therefore, understanding the most harmful events, by modeling how they affect failures' occurrence and duration, is necessary to plan proactive and reactive measures to reduce the damage caused by them, and is a currently highlighted area of study [35, 36, 37, 38].

Regressions are a statistical technique that helps model events of interest. Furthermore, it helps identify and measure the aspects that influence these events. Failures in Electrical Power Distribution Systems can be modeled as the event of interest in Regression models. Factors such as network characteristics and weather events can be included as covariates to check their influence on the occurrence of failures. Generalized Linear Models are a generalization of conventional linear regression and allow for response variables with error distribution models other than the normal distribution, which is a traditional method. It enables the link function that connects a linear model to the response variable and allows the size of the variance of each measurement to be a function of the predicted value.

In [39], this approach was used to model count data to propose a failure rate regression model. The count data is the number of failure occurrences for each day in an observed period. Weather events are included as covariates, using maximum daily values. The included covariates included the number of atmospheric discharges and the maximum occurred wind gust speed. Poisson and Negative Binomial distributions were examined to determine which distribution best fit the failure data. Negative Binomial presented the best result, and the failure rate from the proposed model was defined as in Eq. 3, where nat is the occurred number of atmospheric discharges and the wg is the daily maximum wind gust speed in km/h.

$$\lambda_d = exp(2.2507716 + 0.0011096 nat_i + 0.0274958 wg_i) \tag{3}$$

Another option is to model the time as a random variable. The quantity of failures can be measured with the lifetime of the Distribution Systems. Lifetime is when the system operates in its normal state, i.e., without failures, until the occurrence of a failure event. Similarly, failure (or interruption) durations are another problem that involves time and can be approached by assuming time as a random variable. Survival or Reliability Analysis Regression is a statistical technique that works specifically with problems involving time until the occurrence of an event of interest [40]. Reliability Analysis Regression has three main techniques: non-parametric, parametric, and semi-parametric. For power systems reliability studies, non-parametric and parametric are the most usual choices. Although it has constraints regarding the quality, quantity, and accessibility of data, non-parametric analysis is the technique where fewer assumptions are made, resulting in models closer to the data. For instance, the model cannot provide a probability of the event of interest to occur if the data does not cover the specified time duration. The goal of

parametric techniques is to overcome this aspect by offering a more precise model to accommodate any required range. This difference can be observed in Fig. 4, where the Kaplan-Meier estimator (Non-Parametric) and Weibull Distribution (Parametric) reliability curves are compared.

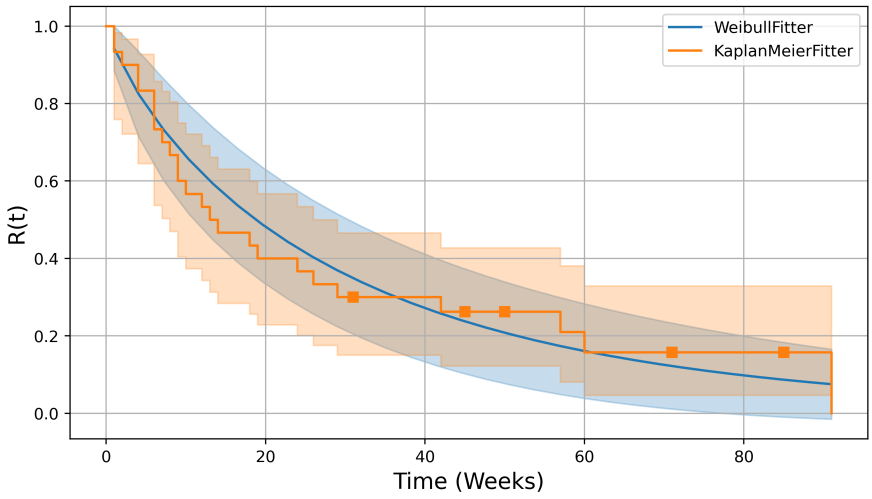


Fig. 4: Reliability Function using Kaplan-Meier Estimator (non-parametric technique) and Weibull distribution (parametric technique) for the same sample of data.

Reliability or Survival Function ($R(t)$) is the probability that a subject survives longer than time t. It is the main output to estimate the lifetime or failure duration. Defining Pr as probability, t as time, and T as a random variable denoting the time of occurrence of the event of interest, the reliability function, $R(t)$, can be defined as in Eq. 4:

$$R(t) = Pr(T > t) \tag{4}$$

Lifetime distribution functions (F) is defined in Eq. 5 as the complement of the reliability function $R(t)$:

$$F(t) = Pr(T \leq t) = 1 - R(t) \tag{5}$$

Selecting a probability distribution is crucial for fitting a parametric reliability model. The primary possibilities from the literature, including the Exponential, Weibull, LogNormal, and LogLogistic distributions, will be assessed for building a model for the time to failure and the time to repair, i.e., interruption duration. Eqs. 6, 7, 8, and 9 present the reliability function for Exponential, Weibull, LogNormal, and LogLogistic distributions respectively.

$$R(t) = exp\left(\frac{-t}{\lambda}\right), \lambda > 0 \tag{6}$$

$$R(t) = exp\left(-\left(\frac{t}{\lambda}\right)^{\rho}\right), \lambda > 0, \rho > 0 \tag{7}$$

$$R(t) = 1 - \Phi\left(\frac{log(t) - \mu}{\sigma}\right), \sigma > 0 \tag{8}$$

$$R(t) = \left(1 + \left(\frac{t}{\alpha}\right)^{\beta}\right)^{-1}, \alpha > 0, \beta > 0 \tag{9}$$

Accelerated Failure Time (AFT) modeling is also used to include covariates. Covariates can be considered on the shape and scale parameters of the models. For example, we can model n covariates influencing the Weibull distribution scale parameter λ. The regression for λ will be as in Eq. 10, and the respective reliability function will be as in 11.

$$\lambda(\mathbf{x}) = exp(\beta_0 + \beta_1 x_1 + ... + \beta_n x_n) \tag{10}$$

$$R(t; \mathbf{x}, y) = exp\left(-\left(\frac{t}{\lambda\,(\mathbf{x})}\right)^{\rho(y)}\right) \tag{11}$$

In [41], the non-parametric Reliability Analysis technique called the Kaplan-Meier estimator was used to estimate the outage duration of different groups of interruption durations for a real Power Distribution System. In the mentioned paper, between the failure causes groups Atmospheric, Environmental, Urban, Operational Needs, and Equipment Failure, Atmospheric causes are the factor that leads to more extensive repair time. In [42], a similar result for these groups was found when modeling the lifetime of Distribution Systems.

As the Reliability Analysis Regression technique allows the modeling of the lifetime and failure durations of power systems, a complete reliability framework can be made, allowing, for example, to sample trustworthy times to failure and duration of such failures, which combined with operational aspects of these systems, can be used to calculate reliability indexes, such as SAIFI and SAIDI.

A detailed analysis was made in [43], explaining how to use Reliability Analysis Regression to model failures in Distribution Systems. Statistical criteria were used to determine the best distribution for the interruption duration model. Of the numerical criteria, the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) stand out. QQ-Plot, a graphical evaluation criterion, is very relevant to decide the best statistical distribution for the proposed model. Failure data from real distribution systems were used to develop such a study.

They found that the LogNormal distribution presented the best result overall. The final equation for the interruption duration for failures in Distribution Systems is shown in Eq. 12.

$$R(t) = 1 - \Phi\left(\frac{log(t) - 3.9110}{1.7851}\right) \tag{12}$$

They also applied the same approach to model the lifetime of Distribution Systems. Weibull distribution presented the best result. Eq. 13 presents the final equation for the lifetime model.

$$R(t) = exp\left(-\left(\frac{t}{127.8349}\right)^{0.5208}\right) \tag{13}$$

After determining the best statistical distribution for each model, covariates were included to evaluate the impact of each event and how it modifies the number of failures and interruption duration. The tests included maximum daily values for six weather events: number of atmospheric discharges, wind speed, precipitation, maximum and minimum temperature, and relative humidity. The first three showed to be the more relevant for the number of failure events.

In Eq. 14 and 15, the reliability function for the interruption duration (t_d) and the lifetime (t_l) of Distribution Systems models with weather covariates $(\mathbf{x}) = [AD,\ WS,\ P,\ T_{\max},\ T_{\min},\ RH]$ were presented. AD represents the number of Atmospheric Discharges, WS is Wind Speed, P is precipitation, $T_{\max}$ is for maximum temperature, $T_{\min}$ is for minimum temperature, and RH is for relative humidity. To use the covariates, wind speed is in kilometers/hour, precipitation is in millimeters, temperature is in Celsius degrees, and relative humidity is in percentual.

$$\begin{aligned}\mu(\mathbf{x}) &= 4.5715 + 0.0001AD + 0.0089WS + 0.0027P - 0.01021T_{\max} \\ &\quad + 0.00636T_{\min} + 0.00125RH \\ R(t_d) &= 1 - \Phi\left(\frac{log(t_d) - \mu(\mathbf{x})}{0.2387}\right)\end{aligned} \tag{14}$$

$$\begin{aligned}\lambda(\mathbf{x}) &= 4.9715 - 0.00064AD - 0.0277WS - 0.0035P + 0.0106T_{\max} \\ &\quad - 0.0123T_{\min} - 0.0017RH \\ R(t_l) &= exp\left(-\left(\frac{t_l}{\lambda(\mathbf{x})}\right)^{-0.4031}\right)\end{aligned} \tag{15}$$

Another example is shown in Figs. 5 and 6. The reliability curves of failure duration and lifetime models presented in Eq. 14 and 15 are demonstrated, with wind speed and the number of atmospheric discharges, which are two relevant factors for the models, assuming different values and the other covariates were used with their average values. For both cases, when the wind speed is slow and the number of atmospheric discharges is low, the chances of the occurrence of the event of interest are below the median case represented by the baseline curve. For the failure duration, it means that the probability of observing a longer interruption is below the baseline curve. For the lifetime, i.e., time to failure, the probability of observing

a longer working time before a failure is above the baseline curve. However, as the combinations of atmospheric events present greater values, the chances of failure duration being longer and the system remaining for less time without a failure occurrence decrease drastically.

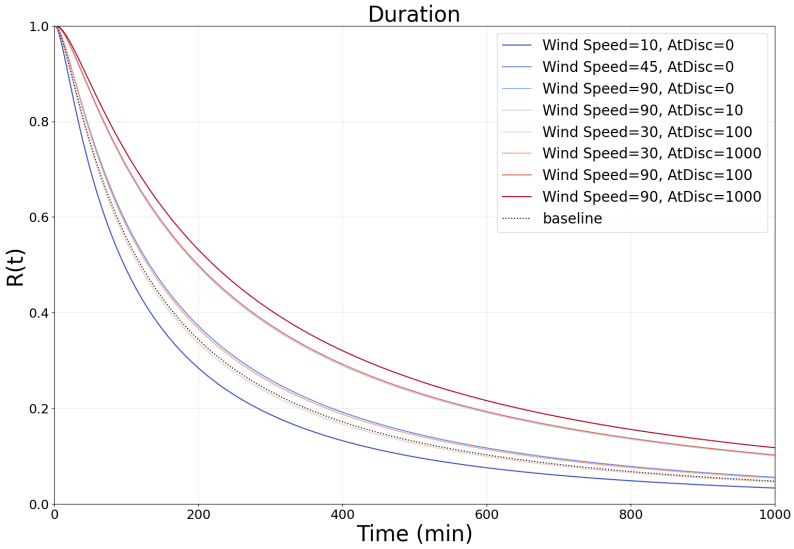


Fig. 5: Failure Duration Model Reliability Curve $R(t_d)$ with weather covariates (Eq. 14). The baseline curve uses the median value from the weather events for the covariates. Wind Speed in kilometers/hour and number of atmospheric discharges are tested with different values.

Depending on the desired application, the proposed covariates can be maintained or removed, and new ones can be added if new data is available to fit the reliability regression models.

3.3 Service Restoration

The representation of the electric power distribution network is $G_E = \{V_E, E_E\}$, with nodes in V_E connected by edges in E_E. Conventional distribution systems maintain a radial layout, i.e., without cycles. After faults occur, there is the necessity to perform service restoration, which consists of operating the normally open and closed switches to reenergize healthy out-of-service consumers and maintain the system's radiality followed by repairing the faulted components. Such operations are performed by field crews that travel through the road network to operate the switches and perform repairs. These switches divide the electrical system into several sectors,

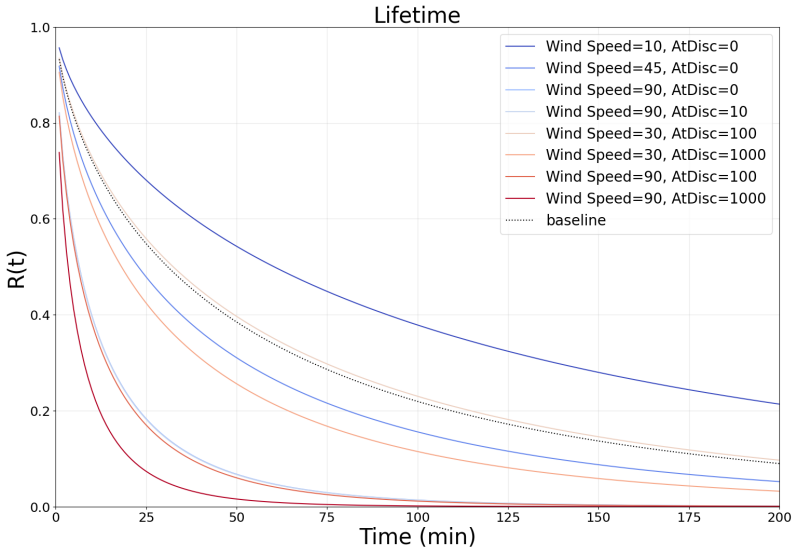


Fig. 6: Lifetime Model Reliability Curve $R(t_l)$ with weather covariates (Eq. 15). The baseline curve uses the median value from the weather events for the covariates. Wind Speed in kilometers/hour and number of atmospheric discharges are tested with different values.

and can also be referred to as node cells [44]. This section presents an approach for routing maintenance teams and obtaining feasible manual switch operations and network reconfiguration during teams' patrols.

3.3.1 Maintenance teams routing

Traditionally, the approach to outage management involves delineating the affected area by cross-referencing data from multiple sources after a system failure and activating protective measures [45]. Subsequently, distribution system operators utilize remotely controlled switches to narrow down the affected area before deploying repair teams to inspect nearby power distribution lines for damage and ensure safety before re-energization [46]. This process necessitates a thorough inspection of the entire affected zone by maintenance crews. Consequently, the patrol method can be viewed as a routing challenge, wherein, in the worst-case scenario, the fault remains undetected, leading crews to inspect the entire affected area.

The displacement of the maintenance teams may be impacted by the state of the road network. Using the notation $G_R = \{V_R, E_R, w_R\}$, the road network can be represented as a directed weighted graph. It has nodes in V_R that represent geographic locations, edges in E_R that represent directed roadways connecting these nodes, and a weight function $w_R : V_R \times V_R \rightarrow \mathbb{R}_{\geq 0}$ that assigns distance values to edges to facilitate

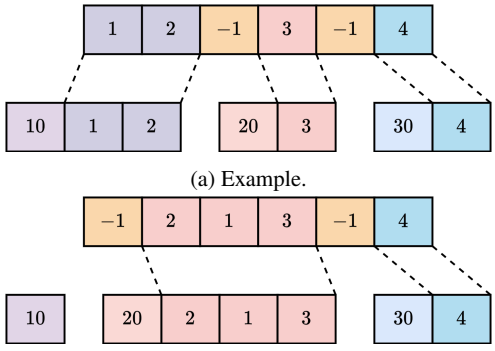
transportation and logistical analyses. For instance, there are various reasons why traffic speeds can vary. The weather, the day of the week, the time of day, and unplanned events are just a few of the many factors that affect traffic flow, according to [47]. Thus, it is crucial to include external environmental information to reduce prediction errors. The study of predicting traffic flow in unforeseen circumstances, such as the absence of real-world data, accidents, and congestion, as well as the applications of integrated models to traffic forecasting are of great interest to the scientific community [48]. Moreover, simulation frameworks can be used to quantify the traffic speed effect on power distribution systems' reliability indices [49].

To perform routing of maintenance teams, we can express a set of nodes $P \subseteq V_G$ that represents the sites to be investigated. First, the power distribution system must be projected towards the road network. This step requires geoprocessing and graph theory techniques. One technique to carry out this projection is to assign each node of the electricity distribution system to the nearest road network node. Another technique involves adhering each electrical node's projection to the nearest edge of the road network by splitting it into two new edges. Before being dispatched to address failure occurrences, maintenance teams must already be displacing, carrying out repairs, or performing some other routine tasks, as opposed to beginning in depots or operational bases as in usual routing problems.

A broader approach involves initiating the patrolling process for the available crews set M at nodes $\mathbf{s} = \{s_m \colon s_m \in V_G\}_{m \in M}$ rather than at operational bases. To utilize this alternative method, specifying all s_m as the operational base location is acceptable. Another relevant element is the delay component τ_m, which signifies the time when the mth team completes its previous task and commences patrolling, denoted by $\boldsymbol{\tau} = \{\tau_m\}_{m \in M}$. Fig. 7a illustrates an appropriate encoding for the patrolling problem and a permutation operator applicable to such representation, for a Genetic Algorithm (GA). In this illustration, the -1 element divides the itineraries of the three available maintenance teams. The nodes to be inspected are labeled as $P = \{1, 2, 3, 4\}$. In addition, teams 1, 2, and 3 begin patrols at different points $\mathbf{s} = \{10, 20, 30\}$. Permutations of such an array, e.g. Fig. 7b, should be used in this situation by the optimization method. By performing permutations, the order of the nodes and the team assigned to the node change. The path to travel from one point to another through the road network graph is obtained through Dijkstra's algorithm.

The element -1 serves as a marker to partition the inspection points for each crew. Through permutation, this approach allows teams to interchange their patrolling regions. Such solution encoding aids in determining the optimal number of teams from the available ones, as some teams may remain inactive. Figure 8 illustrates a restoration scenario where only two of four potential teams were deployed.

Different characteristics, like the distance of the network's edges, $w_G(i, j)$, are incorporated in the edge weight function $\psi \colon V_G \times V_G \to \mathbb{R}_{\geq 0}$, which is specified in Equation (16). Two velocity levels are recognized in this context, v_p and v_d. In particular, when examining a node that is part of the set of necessary nodes in P, v_p is chosen, and v_d is the velocity related to dislocation speed between road network nodes until arriving at another node in P. Additionally, the function $t_v \colon V_G \to \mathbb{R}_{\geq 0}$



(a) Example.

(b) Permutation.

Fig. 7: Illustration of the encoding schema used to perform maintenance teams routing. Three teams are stationed at the starting locations $\mathbf{s} = \{10, 20, 30\}$ in this illustration of the patrol solution. Points for inspection are $P = \{1, 2, 3, 4\}$. In Fig. 7a, the first team's route begins from position 10, followed by positions 1 and 2, respectively. Simultaneously, the second team travels from position 20, which is its initial position, to position 3. Traveling from 30 to 4 is the third team journey. These patrolling are accomplished simultaneously by the three available teams. There is a permutation of this case in Fig. 7b where the second team moves from point 20 to 2, then 1 and 3, while the first team stays in place. The same route is used by the third team.

indicates the time required to operate the manual switches necessary to perform power system restoration.

Another aspect incorporated in Eq. (16) is a factor a_j where components with higher change to be experiencing a failure will have their edge weight reduced to be prioritized during the routing solution. The weights of the paths between the road network nodes are modified to prioritize regions with a high estimated likelihood of containing failures through the factor $a_j = {}^{1}/_{(1+\hat{\lambda}_j)}$, where $\hat{\lambda}_j$ is an estimated failure rate or failure probability for the i-th component, as discussed in Section 3.2. Critical entities such as hospitals, fire departments, information centers, banks, security departments, and industries are accounted for in a critical load factor l_j. As noted in previous works [50, 51], l_j may occasionally integrate another weighting for combining other information sources or priority settings. The χ acts as an indicator function: in the case where $j \in P$, $\chi_P(j) = 1$; in the other case, $\chi_P(j) = 0$.

$$\psi(i, j) = a_j l_j \left[\frac{w_R(i, j)}{(v_p - v_d)\chi_P(j) + v_d} + t_v(j) \right] \tag{16}$$

The routing process's objective function $f(\mathbf{s}, \boldsymbol{\tau}, \mathbf{x})$, which should be minimized, is represented by Eq. (17). First, for a given set of routes $\mathbf{x}$, this equation calculates the cost of each crew member's m to travel through the weighted shortest path using the weighting specified in Equation (16). The length of the weighted shortest path

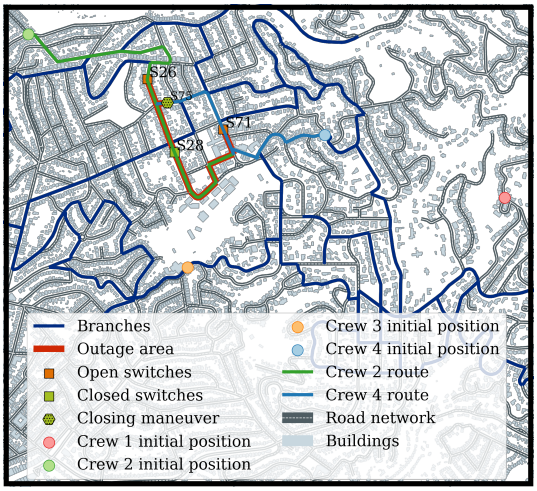


Fig. 8: This example involves a restoration process of a surrogate distribution system having an outage area of 1.18 km and two sectors. Initially, there were 391 out-of-service customers isolated from the remainder of the system by manually operated switches S26, S71, and S75. In addition, there were four available crews. The representation of the patrolling procedure shows the route for teams 2 and 4. These routes comprise their journey and outage area patrolling, chiefly performed by team 2. Finally, while inspecting, team 4 handles S75 closing and restores out-of-service customers in 0.23 hours. During the entire process, teams 1 and 3 remain inactive.

between nodes i and j is calculated by the function $\delta(i, j)$ and using the weights calculated in Eq. (16). For the m-th route, $\delta(s_m, x_{m1})$ represents the first portion of the journey from the starting point in $\mathbf{s} = \{s_m : m \in M\}$ to the first visiting point described in $\mathbf{x}$. Then, the second and third terms take into consideration the order in which node inspections are performed as well as the amount of time needed to complete actions like manually operating a switch. Moreover, the optimization process uses the minimax method [52] to identify and minimize the crews' pathways with the maximum cost, i.e., to minimize the worst-case.

$$f(\mathbf{s}, \boldsymbol{\tau}, \mathbf{x}) = \max_{m \in M} \left\{ \delta(s_m, x_{m1}) + \sum_{j=1}^{|P_m|-1} \delta\left(x_{mj}, x_{mj+1}\right) + \tau_m \right\} \tag{17}$$

3.3.2 Manual switches operation and network reconfiguration during patrol

The majority of distribution systems operate in a radial arrangement, which prevents a load node from being connected to a power source (substation or distributed generation) simultaneously through more than one electrical path. Radiality is the

constraint that needs to be respected when defining switching operations. During service restoration execution, the network topology is changed by the maneuver of switches (changes from open to closed states and vice-versa) to reconnect healthy out-of-service customers and repaired elements. Since switches are the components responsible for topology changes, they define the minimum level of granularity for network operations. The smallest set of network components that can be isolated from the rest of the distribution system by opening switches or protection devices is called a sector. Fig. 9 shows a sector representation.

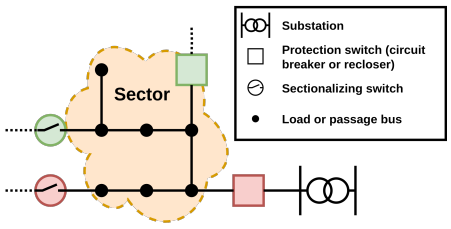
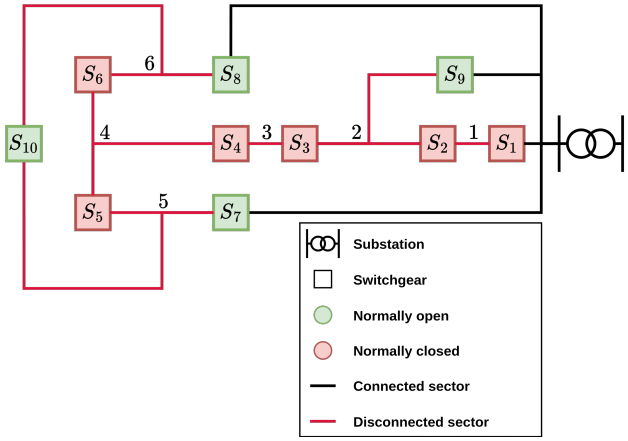


Fig. 9: Illustration of a sector of the distribution system. The set of components that can be isolated from the rest of the distribution system by opening switches or protection devices defines a sector.

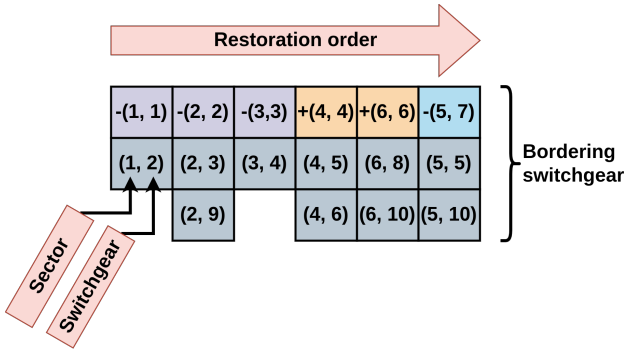
After faults occur, protection devices' action and isolation operations result in disconnected components. These components are divided into sectors, and a candidate restoration solution is encoded in a matrix $\mathbf{Y}$. In essence, each element of $\mathbf{Y}$ is a signed pair, and each pair comprises the number of a sector and the number of the switch that must be closed to restore service to this sector y_{ij}. The operation that reenergizes a sector is defined by closing the switch that connects a de-energized sector to an energized one, and these operations are represented as the second value of each signed pair in the first line of the matrix ($y_{1j},\ \forall j$). The maintenance teams must patrol the sector and open the other bordering switch before the closing maneuver to address the radiality constraint. These open maneuvers are represented in the next lines of the matrix $\mathbf{Y}$, and each $y_{ij},\ \forall i > 1$ indicates the necessary switch open maneuver to restore the j-th sector. In this situation, the sequence of the sectors in this encoding corresponds to the sequence of the restoration procedure.

An advantage of this encoding is that it allows a reduction of the number of maneuvers to restore adjacent sectors by combining the switch operations. This is achieved by combining the signal of sectors, where adjacent sectors with the same signal compose a super-sector that can be restored through a combination of their open and close switch operations, the open maneuvers of each sector can be skipped due to the need to close the same switch for the restoration of the next sector in the super-sector. Fig. 10b depicts an example of a reconfiguration encoding illustrating the number of switch operations by composing super-sectors. Given a feasible reconfiguration to achieve service restoration encoded in $\mathbf{Y}$, the routing

optimization algorithm from Section 3.3.1 must be executed to guarantee the teams take an optimized route.



(a) System.



(b) Encoding.

Fig. 10: A distribution system with several disconnected sectors, sectionalizing equipment, and a substation is shown in Fig. 10a. The encoding maps out the disconnected area into an array of its sectors. Fig. 10b illustrates one possible network reconfiguration solution encoding. Each tuple in the array has two elements: the first is connected to the sector being restored and the second is related to the switch being closed to restore it. The other bordering switches must be open before the closing maneuver to maintain radiality during service restoration. By the signal change, it is possible to combine adjacent sectors into a single restoration stage using this encoding. The example generates three super-sectors and three stages of restoration, the first of which is composed of sectors $\{1, 2, 3\}$. Moreover, the closing switch is S_1, and the opening switches are S_4 and S_9. $\{4, 6\}$ is the second super-sector with the closing switch as S_4, while the opening switches are S_8, S_5, and S_{10}. The closing switch is S_7 for the last super-sector, 5, with the opening switches S_5 and S_{10}.

This encoding ensures that one switch will be closed in each restoration phase, however, it does not guarantee practical solutions. Then, depending on the following variables, Equation (18) evaluates the practical feasibility of the solution, and n_o is the number of elements leading to an unfeasible solution.

- o: The solution is impractical.
- p: The switch being closed is a neighbor of nodes supplied with electricity.
- q: The switch being closed must intersect another super-sector already in service or previously restored.
- r: the elements of a super-sector are contiguous in the graph
- s: a closing switch can only be used to restore no more than one super-sector

$$((\neg p) \wedge (\neg q)) \vee (\neg r) \vee (\neg s): \implies o \tag{18}$$

The objective function to be minimized to obtain a service restoration $\mathbf{Y}$ is presented in Eq. (19). For every restoration step $\iota \in I$, where I represents the whole restoration process, the teams' starting positions $\mathbf{s}_\iota$ and the delays associated with them $\boldsymbol{\tau}_\iota$ are updated iteratively. Furthermore, the penalty factor ρ is increased by the number of feasibility infractions n_o to account for violations. In this case, a routing solution is represented by $\mathbf{x}_\iota$, where P_ι is the set of all nodes, including switches, in the super-sector ι. In Section 3.3.1, the optimal routing solution is represented by the term $f^*(\cdot)$, which comes from Equation (17). The count of customers that are still out-of-service is then multiplied by this solution. This count is represented by the summation of the customers c_ζ within super-sector ζ during the ιth restoration step.

$$g(\mathbf{Y}) = \sum_{\iota \in I} f^*(\mathbf{s}_\iota, \boldsymbol{\tau}_\iota, \mathbf{x}_\iota) \sum_{\zeta \in I:\, \zeta \geq \iota} c_\zeta + \rho n_o(\mathbf{Y}) \tag{19}$$

This approach consists of a bi-level optimization procedure, each service restoration candidate solution $\mathbf{Y}$ aims to minimize $g(\mathbf{Y})$, and the teams routing necessary to perform the service restoration are obtained by minimizing the $f(\cdot)$ (Eq. 17). A detailed formulation and discussion of such an approach is presented in [49].

A tournament-based genetic algorithm (GA) was utilized for both levels of optimization. In the lower level, which is the routing mechanism in Subsection 3.3.1, the GA employs ordered crossover (OX1) and swap operators as crossover and mutation, respectively. These operators generate controlled permutations of genes, which are the nodes to be inspected during the patrolling phase. The chromosomes are the routing solutions in this case. In the upper level, the crossover operator is maintained to produce permutations among sectors. However, different mutation operators are used in this stage. The swap operator is employed to exchange the order of the sectors and which switch will energize the sector, in addition to the flip-bit for signal change. In this case, the swap mutation is performed in the horizontal and vertical directions in Fig. 10b. Readers who want more information about these genetic operators should refer to [53]. The GA parameters used to produce the results presented in Section 4 are as follows:

- Population size: 300
- Crossover rate: 0.8
- Mutation rate: 0.2
- Tournament size: 3

3.4 Resilience Assessment using DBN

To assess the resilience of systems using a DBN, it is necessary to define the DBN structure and conditional probabilities to allow simulation of disruptive events effects through evidence propagation followed by the resilience estimation. These aspects will be detailed in the remainder of this Subsection.

3.4.1 Structure of the DBN

The first step in constructing the DBN for resilience assessment is to obtain its structure. It involves defining the representation of each node within the DBN, and in this method, every node is conceptualized as a component within the topology of a DS. While this approach can be applied to model various CIs, the primary focus here is on power systems. Consequently, each node corresponds to an electrical bus, which can be either a load or a feeder. Denoted as N_t, the set of nodes for the DBN's time slice at time t is established. Figure 11 a) shows the topology of a typical electrical power system, which is then used to model a DBN's structure, as exemplified in Figure 11 c).

Each node is modeled as a binary variable, offering two possible values: $n_t = 0$ signifies that the n-th node (where $n \in N$) at time t is functioning normally, and $n_t = 1$ indicates that the node is either out of service (de-energized) or in a state of failure. It's essential to emphasize that with this strategy, every component within the energy system will correspond to a dedicated DBN node at each time slice, as exemplified in Figure 11 c). Consequently, the process of assigning a node a specific value (either 0 or 1) is employed to simulate the occurrence of a specific component failure, as shown in Figure 11 b). This simulation is later utilized in conjunction with evidence propagation to model the probability of disruptive events.

3.4.2 Conditional Probability Tables

After the structure of the DBN is defined, the next step is to model the CPTs that will define the statistical dependencies of each node. To do so, two reliability models are necessary. The model of failure occurrence probability given a time instant $(1-R(t_l))$ and the model of failure duration probability $(R(t_d))$. Subsection 3.2 presented and detailed these two models. In this Subsection, we will denote for a given time instant t that $1 - R(t_l = t) = f(t)$ and $R(t_d = t) = r(t)$.

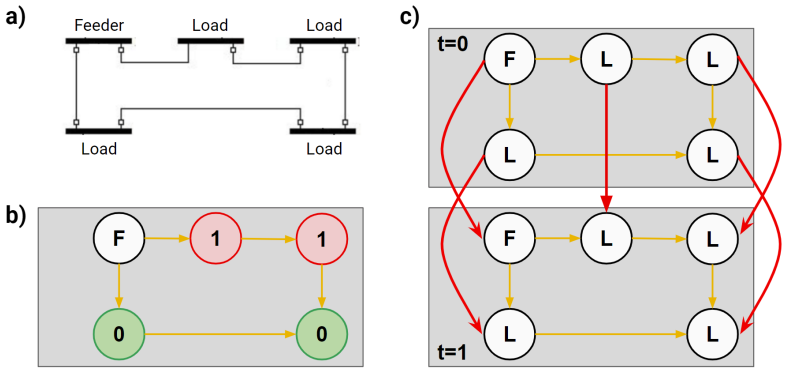


Fig. 11: Procedure to define the DBN's structure based on the energy system's topology. In a), the original topology of a modified 5-bus test case is presented. In b), the respective static BN is shown, with a typical failure scenario (green nodes working; red nodes in failure). In c) a DBN with replicated static time slices is modeled (intratemporal edges are yellow, intertemporal are red).

To model the CPTs of all nodes in the DBN, the following rules are applied:

1. All feeders start in normal operational conditions (probability of failure equal to 0 at $t = 0$);
2. If all feeders connected to a given node n at time t are in failure, then the probability of the node being in failure at the same time is equal to 1, which is mathematically modeled as $P(L_t = 1)$;
3. If a load node L has been in failure for the previous k time slices (i.e $L_{t_i} = 1$ for $t_i \in \{t-1, t-2, ..., t-k\}$) then the probability that such a node goes back to normal operational conditions in the next time slice t is mathematically expressed by $r(t)$;
4. If a load node L has been in normal operational conditions for the previous k time slices (i.e $L_{t_i} = 0$ for $t_i \in \{t-1, t-2, ..., t-k\}$) then the probability that such a node experiences a failure in the next time slice t is mathematically expressed by $f(t)$.

With the aforementioned rules as a starting point, Figure 12 shows an example. It is important to highlight that while the Figure shows a simple example with only two nodes (1 feeder and 1 load) with the proposed method, it can model any radial energy system with multiple feeders and loads. Additionally, different topologies (depending on the action of protective devices or islanding mode) can be assessed by this approach, by adapting the final DBN accordingly. This allows the proposed method to be flexible to deal with distributed generation in active Power DSs.

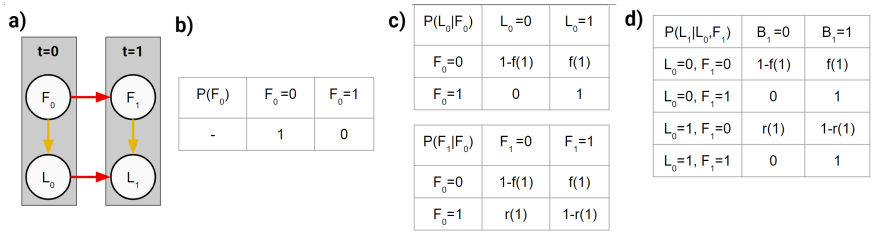


Fig. 12: Example of a simple energy system with a single feeder connecting a single node. In a), a DBN's structure with two-time slices is presented. In b), we have the CPT for the root node. In c), two possible CPTs for a child node, are conditioned to its parent. In d), a CPT of a child node with two parents.

3.4.3 Simulating Disruptive Events with Evidence Propagation

With the provided tools, we can already model a full DBN, consisting of both its structure (nodes and edges) as well as the conditional probability tables relating different nodes. With this in mind, the next step is to use the process of evidence propagation to properly simulate failure scenarios in a DS. In the proposed approach, each node in the DBN is related to a physical component (in our context, an electrical bus, either a load or a feeder). With this in mind, consider that a disruptive event consists of defining a set of components, i.e., DBN nodes, to be in failure for a specific period. By remembering that each DBN's time slice represents a period in time, and by properly defining the temporal scale (the time that passes between subsequent time slices), the following procedure can be used to define a disruptive event:

1. Consider a set of nodes to be in failure ($N_f \subset N$), each one with an instant of failure (t_f) and a failure duration (t_d);
2. Consider all time slices in the DBN ($t_s \in \{0, ..., T_h\}$), where T_h is the time horizon, representing the maximum number of time slices;
3. Consider that the state of a given DBN's node (0 for working; 1 for failure) at a given time t_s is represented by $s_{n,t}$;
4. For each node in failure: if $t_f \leq t_s \leq t_r$, the node's state is considered as in failure ($s_{n,t_s} = 1$); otherwise, it is in working state ($s_{n,t_s} = 0$).

To illustrate the procedure, Figure 13 a) presents a simple example of a 5-bus energy system with a specific disruptive event. Figures 13 b), c), and d) depict how the DBN assigns values to its nodes based on the failure scenario. It's noticeable that while a subset of nodes has fixed values (either 1 or 0, depending on the disruptive event), all other nodes (depicted in blue in the figure) don't receive fixed values since they aren't part of the disruptive event. Instead, their Conditional Probability Tables (CPTs) are updated to account for the occurrence of the disruptive event. The process of assigning fixed values to some nodes and updating the probabilities of others is achieved through evidence propagation.

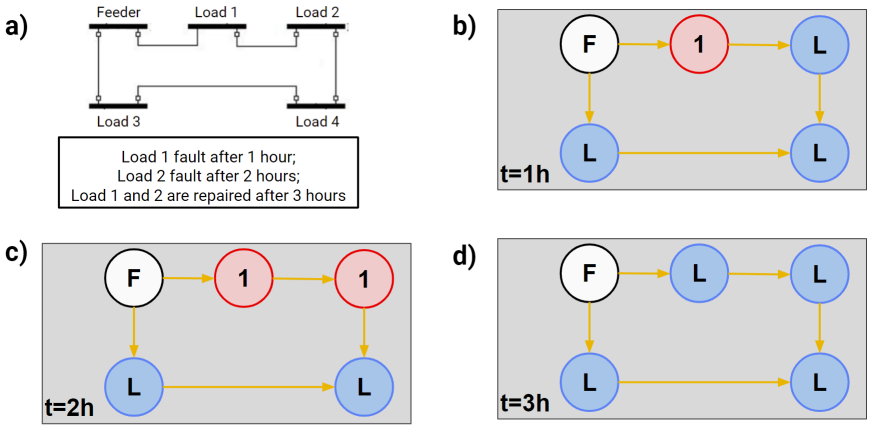


Fig. 13: Figure showing the possibility of modeling disruptive events by using evidence propagation. In a), the 5-bus IEEE test case is presented as a reference, along with a disruptive event (black box). In b), the DBN for $t = 1$ hour is shown, with Load in failure (state equal to 1); in c), for $t = 2$ hours, both Load 1 and 2 are in fault; finally, in d), after $t = 3$ hours, both loads are repaired. Nodes in red are in failure; nodes in blue have no defined state.

3.4.4 Resilience estimation of a DS

After the evidence propagation to simulate a disruptive event scenario, the DBN with updated probabilities can be used to assess the overall Resilience of the energy system. However, since the DBN deals only with probabilities (which represent how likely a component is to be in failure or working state at a given time slice), it is not possible to directly assess a single resilience curve for the whole time horizon. Instead, two possibilities are possible:

1. It is possible to sample from DBNs given the updated CPTs [23], which can be done to simulate various trials by Monte Carlo approaches, and consequently, generate several resilience estimations;
2. With the updated probabilities, it is possible to find the most likely state of the DBN as a whole (in other words: if a component is working or in failure for each time in the time horizon), which is done by the process called Maximum a Posteriori (MAP) estimation [22].

While the strength of applying Monte Carlo approaches to the process of random sampling in DSs has been highlighted previously in the literature [18], the results presented in this Chapter will apply the MAP technique to calculate the most probable state of each component at a given time. Consequently, it is possible to evaluate the most probable resilience curve, by calculating the performance function for each time. The Supplied Energy (SE) will be used as the performance function, which can be calculated as shown in Equation 20, where $P(n,t)$ is the active power demand

for node n at time t. In this case, each value of $s_{n,t}$ will be defined by the process of MAP.

$$SE(t) = \sum_{n \in N} P(n,t) \cdot s_{n,t}, \forall t \in \{0, ..., T_h\} \tag{20}$$

To consider the influence of the disruptive event on the probabilities of the DBN and into the final resilience curve, the 5-bus IEEE test case, shown in Figure 13 a), is used for the simulations presented hereafter. The reliability functions for the CPTs are fixed in $f(t) = 0.1$ and $r(t) = 0.1$, as typical values from the literature [22]. Finally, 3 distinct disruptive event scenarios are considered, as presented in Table 1. Scenario 1 and 2 show the same element being in failure (Load 3), but with a higher time to repair for Scenario 2. Additionally, Scenario 3 has the same time to repair as Scenario 2, but with more elements being in failure (both Feeder and Load 3). With this, it is possible to verify the influence of the time to repair itself (duration of the failure) and the number of components being in failure (intensity of the disruptive event).

Table 1: 3 possible scenarios representing disruptive events, which are simulated in the 5-bus IEEE test case, shown in Figure 13 a). For each scenario, the set of nodes to fail, the failure instant and the repair time are defined.

Scenario number	**Node to failure**	**Time to failure**	**Time to repair**
Scenario 1	Load 3	4 hours	3 hours
Scenario 2	Load 3	4 hours	6 hours
Scenario 3	Feeder and Load 3	4 hours	6 hours

Figure 14 illustrates the evolution of Supplied Energy for the 5-bus test case across the different simulated scenarios. Given that the time to failure is equal for all nodes (occurring at $t = 4$ hours), the decline in Supplied Energy is synchronized. In Scenarios 1 and 2, only a portion of the energy supply is disrupted since a single node goes to a failure state. Conversely, in Scenario 3, where the entire feeder is compromised, the energy supply to all nodes ceases, resulting in a system collapse, i.e., $SE(t = 4\,\text{hours}) = 0$. This showcases the method's capability to model disruptive events of varying severity.

Furthermore, in Figure 14, when comparing Scenarios 1 and 2, we observe that despite both scenarios experiencing the same initial performance decline (attributed to the failure of Load 3), Scenario 2 has a longer period of partial energy supply due to the modeled disruptive event having a 6-hour repair time. In contrast, Scenario 1 features a shorter 3-hour repair time, allowing it to return to normal conditions more swiftly. This demonstrates the DBN's capacity to represent disruptive events with differing durations.

Lastly, it is important to emphasize the DBN's capability to model not only the most probable scenarios (utilizing the concept of MAP) but also to dynamically recalculate the probabilities of a specific component experiencing failure at various

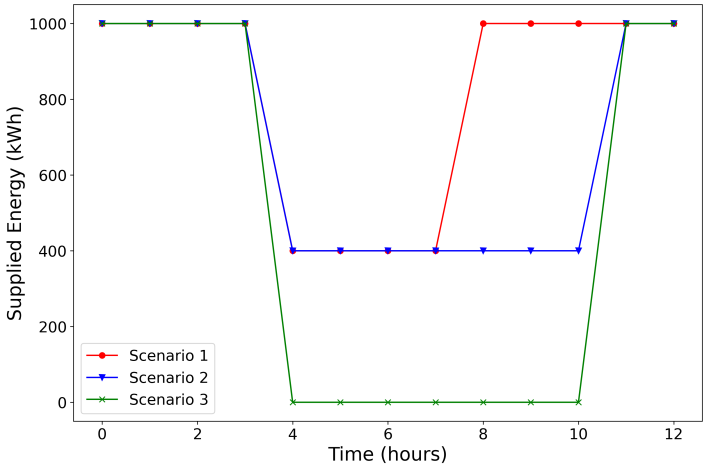


Fig. 14: Evolution of the Supplied Energy of the 5-bus IEEE test case, shown in Figure 13. Each scenario presented in Table 1 is simulated, by applying evidence propagation coupled with MAP, to define the most probable resilience curve in each case.

time intervals, taking into account the simulated disruptive event. The updated Conditional Probability Tables (CPTs) of the DBN, computed through evidence propagation, serve to verify these probabilities, as illustrated in Table 2.

The DBN can reevaluate the probability that a node will be in a failure state at future time points (e.g., $t = 6$, $t = 9$, and $t = 12$) compared to initial evidence from previous times. All failures commence at $t = 4$, as depicted in Table 1. Notably, it becomes evident that the probability of Load 4 being in a failure state increases with the severity of the scenarios: from Scenario 1 to Scenario 2, where the time to repair is extended, and from Scenario 2 to Scenario 3, where the number of components in failure is greater. This is anticipated since both the feeder and Load 3 are components along the electrical path to Load 4 (as shown in Figure 11). This feature facilitates sensitivity analysis of the system's failure response, allowing for an assessment not only of the overall probability trend (whether it increases or not) but also its magnitude (i.e., the numerical change in the final probability).

4 Case Study

In order to allow a better comprehension of the impact of the proposed Multilayer Resilience Assessment, which integrates the models and algorithms presented in Section 3, it will be applied to a test case in which a power distribution system was built upon a real road network, a summary of the test case data is presented in

Table 2: Failure probability of Load 4 over different time intervals, with varying simulated Scenarios (Table 1 presents each scenario separately), illustrates the impact of evidence propagation on updating probabilities in each scenario.

Scenario number	**Probability that Load 4 is in failure at t=6 hours**	**Probability that Load 4 is in failure at t=9 hours**	**Probability that Load 4 is in failure at t=12 hours**
Scenario 1	0.724	0.729	0.699
Scenario 2	0.744	0.756	0.737
Scenario 3	1.000	1.000	0.878

Table 3. The system's electrical and geographical features are available in [54], and more details about it can be found in [49].

Table 3: Summary of the test case used in this case study, the data is available in [54].

Feature	Quantity
Substations	2
Feeders	4
Load nodes	68
Switches	46
Sectors	37
Customers	4697
Road network nodes	928
Road network edges	2309

The initial step of the simulation involves assessing the failure probabilities of all sectors within the test case by utilizing the DBN, where each sector is represented as a distinct node. To this end, the baseline curves presented in Figs. 5 and 6 were used for the repair duration (or failure duration) and time to failure occurrence models, respectively. These models provide the failure and repair probabilities, $f(t)$ and $r(t)$, which are processed by the DBN and generate a conditional failure probability for each sector. Figure 15 illustrates such failure probabilities across the entire system. Notably, the DBN's capability to consider the system's granularity enables the calculation of distinct failure probabilities for each sector. This granularity proves instrumental in informing the restoration algorithm, allowing sectors with higher failure probabilities to be prioritized during restoration efforts. Such prioritization ensures a strategic approach to system restoration, enhancing overall efficiency and resilience.

The variability in the probabilities of failure, as illustrated in Figure 15, is closely tied to the structural arrangement of the DBN. The nodes' probability of failure is determined by the failure duration and time to occurrence models fitted to historical data. However, nodes with multiple paths to the feeder have a lower chance of power interruption, as the loss of power along the energized path can be restored by alternative paths. Importantly, these probabilities represent prior probabilities, capturing the likelihood of failure when no other sections are compromised. These

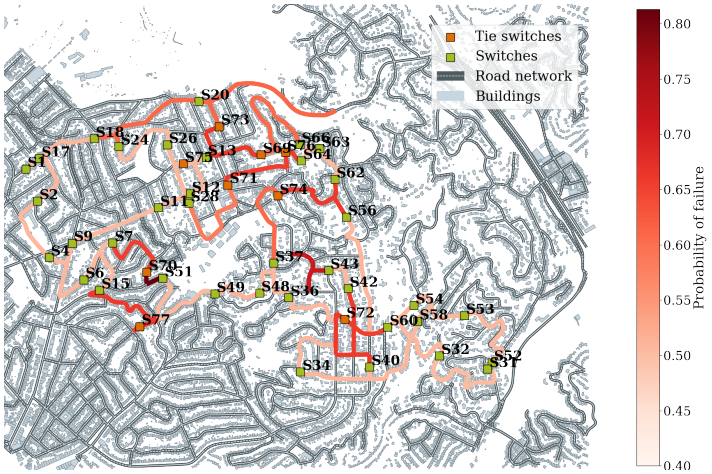


Fig. 15: Spatial distribution of the probabilities of failure engendered by the DBN builder using the failure duration and time to failure models.

initial probabilities of failure provide a baseline for simulating individual points of failure within the system and guiding the restoration process effectively.

Fig. 16 illustrates a fault case with six out-of-service sectors affecting 987 customers, which is more than 20% of the total customers served by the distribution system. There are several ways to patrol and restore power to the out-of-service customers. One possible solution would be to patrol and restore power to one sector at a time. For example, opening device S18 and closing S17 would restore power to the first sector after patrolling it. Then, the sector between switches S20, S18, and S24 could be patrolled and restored, and this process goes on until a fault is found and repaired or all the sectors are restored. However, managing so many switches can not be the better solution due to dislocation and patrolling times summed with the operation time of all the switches, which involves isolating traffic with signalization and preparing their security apparatus before climbing the pole to operate each device.

Another solution would be to patrol the entire outage area and close switch S17 after clearing any fault. This patrolling strategy is widely used by utilities but can also be time-consuming, especially in rural areas. An optimized strategy would combine the best of both worlds: patrolling several sectors at once and operating fewer devices. The service restoration described in Subsection 3.3 provides such optimization from the patrolling perspective by uniting nearby sectors into a single restoration stage. The multilayer approach enables a resilience assessment considering such efficient solutions and is illustrated in the conditions outlined in Fig. 16.

The optimized restoration process, which is considered during the resilience estimation, is executed in two stages. Fig. 17 illustrates the first stage of the service restoration with two crews. In the first phase, crew 1 patrols the lower sections of the

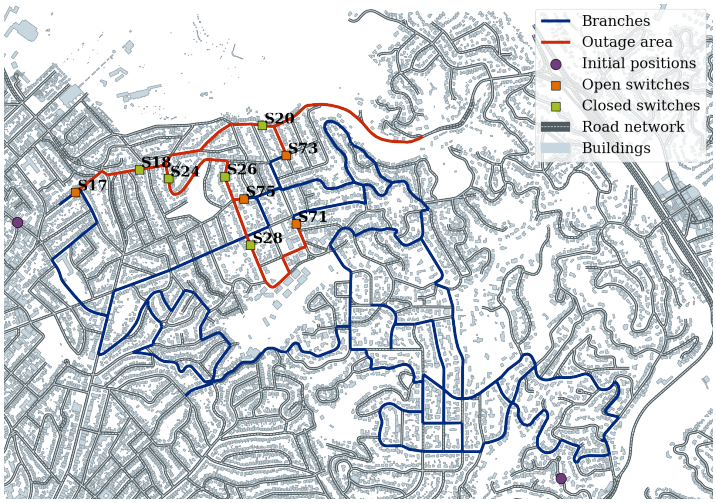
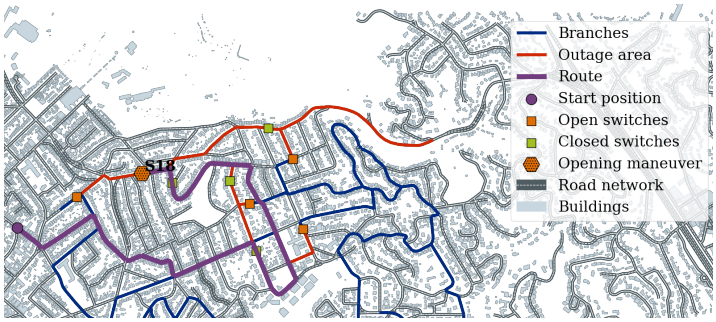


Fig. 16: Illustrated as a spatial depiction of the outage area consisting of six sectors, the initial positions of the two available crews were randomly selected from the road network using a uniform distribution. Initially, the outage area is isolated due to an event-triggering device S17. Furthermore, devices S71, S73, and S75 are tie switches capable of transferring the outage area to another feeder and restoring all the unserved customers. This situation includes a count of 987 disconnected customers.

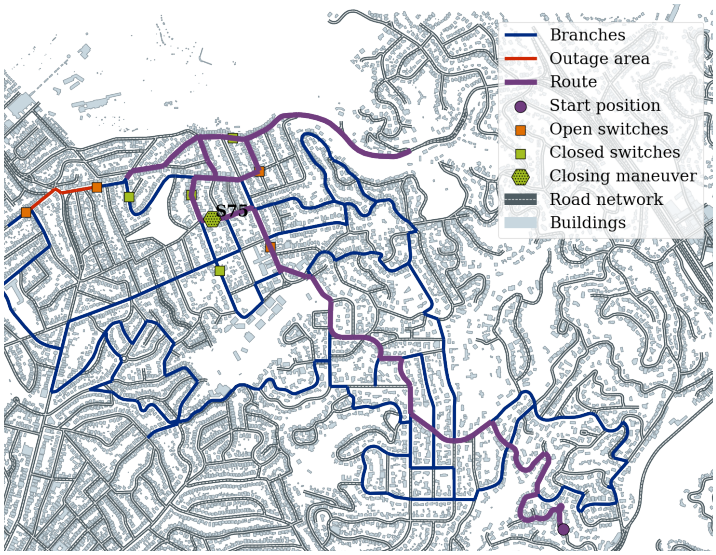
network, relocates to device S18, and opens it, effectively isolating this sector from the broader network. Meanwhile, the second crew inspected the remaining nodes within this network and closed the tie switch S75, restoring power to 934 customers in 26.4 minutes. The second stage is presented in Fig. 18. In the final stage of power restoration, crew 1 is assigned the task of maneuvering the sectionalizing device S18 to restore power to the remaining sector, which has the last 53 de-energized customers.

This example incorporates the complete approach for patrolling optimization, i.e., the bi-level optimization algorithm from Subection 3.3 and the probabilities of failure generated by the DBN by using the failure duration and time to failure models from SubSection 3.2 and shown in Fig. 15. In the outage area, a sector experiences the least service interruptions. As depicted in Figure 15, this sector is between switches S17 and S18. As a result, it is given lower priority during patrol and restoration. In the final stage of restoration, this sector is shown being restored in Figure 18. This is an intriguing result if compared to the approach commonly used by utilities, which involves scanning the outage area in the downstream direction by following the topological sort. Such an empirical approach employed by the utilities would start the service restoration in this sector, making the network less resilient.

Besides integrating the optimal service restoration process during resilience assessment, another crucial aspect of the proposed multilayer approach is its capability



(a) Stage 1 - Crew 1.



(b) Stage 1 - Crew 2.

Fig. 17: First stage of the optimized restoration process for the case in Fig. 16. Crew 1 patrols a portion of the network before advancing to S18 and opening it, separating this sector from the rest of the network. After inspecting the complementary nodes in the outage area, crew 2 closes S75 to restore power to most de-energized customers.

to assess the system's resilience across various weather conditions. This evaluation relies on the Reliability Curves depicted in Figures 5 and 6, which illustrate failure duration and time to failure occurrence, respectively. Three distinct weather scenarios are examined: 'Wind Speed=10 km/h, AtDisc=0' (mild weather); 'Wind Speed=45 km/h, AtDisc=100' (moderate severity weather); and 'Wind Speed=90 km/h, AtDisc=1000' (extreme severity weather). A unique DBN is constructed for each weather condition, incorporating differing failure duration and lifetime distributions. Specifically, the failure duration model informs the assessment of the

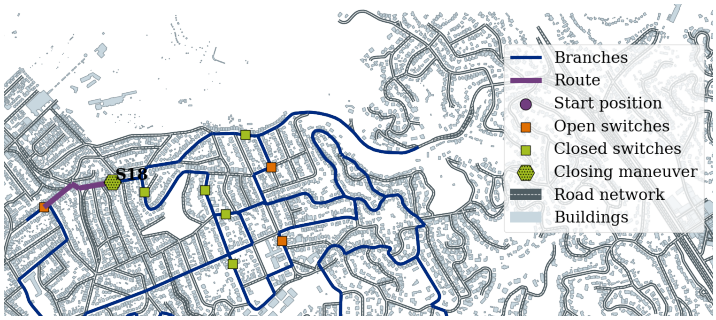


Fig. 18: Second stage of the optimized restoration process for the case in Fig. 16. After the switch maneuvers at stage 1, crew 1 patrols the remaining outage area in the second stage and reenergizes it by shutting device S18. In this last step, crew 2 remains idle and therefore is released for another occurrence.

interruption of a failure for each power system sector, while the lifetime is utilized to determine the failure probability. The probability of failure, in turn, serves as a weighting factor in the routing algorithm, prioritizing nodes with the highest probabilities for visitation. Moreover, the duration is utilized to calculate the duration of the repair crew's stay at each node before proceeding to the next.

Following the application of the service restoration algorithm, it becomes feasible to analyze the sectors that remained disconnected and for what duration. Figure 19 illustrates the progression of the supplied energy over time, incorporating the two-step restoration process outlined in this section and considering the effects of the weather scenarios described in the previous paragraph. Initially, at the onset of the simulation, all nodes are operational, resulting in 100% supplied energy. The restoration process comprises two distinct steps, with the duration of each step influenced by weather variables. Harsher weather conditions lead to longer average failure duration, as depicted in the resilience curve. Consequently, during severe weather events, the system takes more time to return to its normal operational state (supplied energy of 1). For instance, a scenario of low severity (blue curve) requires approximately 100 minutes for full restoration, while extreme severity conditions (green curve) entail nearly 1000 minutes.

In reference to Figure 19, it is essential to consider that, although the total duration for complete system restoration (achieved when supplied energy reaches 1) varied across different weather conditions, the initial drop in supplied energy (approximately to 0.86) and the state of partial repair (with supplied energy at 0.94) remained consistent across all three weather scenarios. This consistency may appear unusual; however, it stems from the simulation procedure, wherein the sector with the highest failure rate—mode of the spatial distribution—was deliberately chosen to experience a failure state. Despite variations in weather conditions affecting the probability of failure, they do not alter the identity of the sector with the highest failure probability relative to others. Consequently, irrespective of the weather conditions, the same

sector experiences failure. Hence, while the timing of repair steps differs depending on the weather conditions, they yield the same performance drop value due to the uniformity in the sector experiencing failure across all evaluated scenarios. As a result, the maintenance teams isolated the affected sector and brought power back to the others before re-energizing the sector between the switches S28 and S71, which was the source of the incident. In this case, the switching operations produced discontinuities in the supplied energy curves.

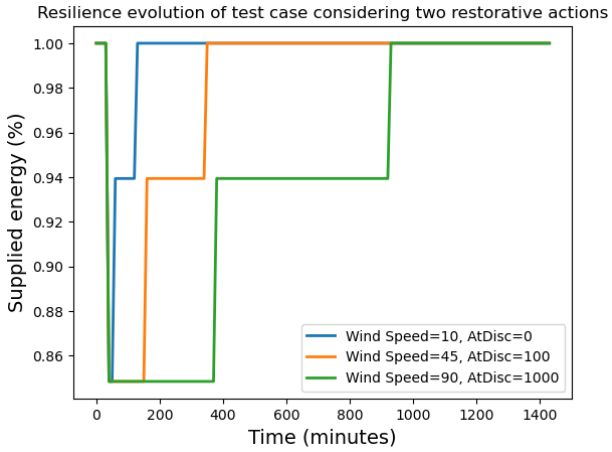


Fig. 19: Resilience over time for the restoration process conducted in the case study. The performance function shown is Supplied Energy.

Such findings underscore the proposed approach's capability not only to statistically model weather variables within reliability curves but also to translate these curves into resilience assessment within a power system, accounting for the service restoration process. Such capability produces specific results that are different from the ones illustrated in Subsection 3.4, where only repair was considered. The main differences are related to the weather effect combined with the service restoration in the performance estimation by evidence propagation and MAP inference. From the results in Fig. 19, it is obvious that the system resilience is lower as the weather severity increases, and the effect of service restoration is clear due to the partial performance bounce back.

It is worth noting that the supplied energy levels depicted in Figure 19 are modeled using the DBN, where the evidence propagation process determines the operational status of each node at every time interval. To achieve this, a 10-minute time interval was employed for modeling the DBN, providing a finer granularity of the repair process, which typically spans around 60 minutes. The flexibility in choosing the length of the time interval, a parameter set by the user, enables the evaluation of resilience even in scenarios involving swift restorations, such as the one depicted,

occurring within an hour. Moreover, this approach accommodates operations that may extend over hours or even days, ensuring comprehensive resilience assessment across varying timescales and conditions.

5 Conclusion

This chapter presented a multilayer approach to assessing the resilience of power distribution systems by integrating different data sources, models, and algorithms. The presented approach integrates reliability regression models to failure occurrence instant and failure duration that take into account for weather variables' effects, a service restoration algorithm to obtain an optimal restoration strategy, and field crew routing, which minimizes the time to service restoration through manual switches operation, and a DBN resilience assessment that uses the reliability regression models and the service restoration algorithm to estimate resilience accounting for all the modeled factors. The integration of these models and algorithms in the multilayer approach is illustrated by a case study with detailed information on the optimized service restoration and the effects on the estimated resilience obtained from the DBN.

Reliability engineering regression is used to evaluate two main measures in the context of power systems: the time between failures and the failure duration. Then, the probabilities related to these two times were used as inputs to the DBN method, where the interactions between different components in the power system are modeled through the conditional probabilities evaluated by the DBN. Unlike traditional approaches, the proposed technique employs DBNs to model the intricate interactions among diverse components within the power system. By utilizing conditional probabilities computed by the DBN, we gain a comprehensive understanding of how different elements influence each other's behavior and contribute to the overall system response to failures. In addition, a service restoration algorithm was used to allow the use of redundant electrical paths, related to the operation of normally open switches, which is mandatory for a better response to failure occurrences.

Moreover, the versatility of DBNs extends beyond mere modeling; they serve as a powerful tool for resilience assessment. Through MAP inference, combined with evidence propagation techniques, the DBN enables us to evaluate the system's resilience under various scenarios. This dynamic approach allows us to simulate and analyze the system's response to different stressors, such as component failures or external disturbances. By propagating evidence of failures through the network, we can quantify the impact on system performance and identify critical points of vulnerability.

The illustration of the methods separately and the case study with their integration allowed us to show how they affect the resilience estimation, the weather severity is mandatory in the rapidity to recovery all the system performance, and the service restoration is fundamental to exploring the redundancies due to the presence of normally open switches present in power distribution systems. The consideration of

all these aspects is fundamental to accurately obtaining a performance estimation, and consequently, the associated system resilience estimation.

This chapter does not explore all the integration possibilities between the presented models and algorithms. Moreover, other aspects can be relevant to a specific power distribution system, such as other weather variables, and specificities related to the service restoration, for example remotely controlled switches and distributed generation, or even to another critical infrastructure that will demand specific adaptations of the multilayer resilience assessment approach presented here. Furthermore, the presented method can be extended to consider the interdependencies between different systems during resilience assessment, and the resilience estimation can be performed by simulating various trials of the DBN by Monte Carlo approaches instead of using the MAP inference.

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